

Qn	Suggested Solutions
5(a)	The resistance R of a circuit component is defined as the ratio of potential difference V across the conductor to current I flowing through the conductor.
(b)	current in series circuit $I = \frac{E}{R_{total}}$ $= \frac{E}{A + B}$ pd across $A = IA$ $= \left(\frac{E}{A + B} \right) A$ $= \frac{A}{A + B} E \text{ (Shown)}$
(c)(i)	As the temperature of the thermistor is raised, its resistance will decrease.
	This will result in the combined resistance of the thermistor and the 4000Ω resistance to decrease due to their parallel circuit arrangement.
	As a result, based on the potential divider principle, the potential difference across the 1500Ω resistor will increase and hence, the voltmeter reading will increase.
(ii)	$\frac{1}{R} = \frac{1}{B} + \frac{1}{R_T}$ $\frac{1}{R} = \frac{1}{4000} + \frac{1}{2700}$ $R = 1612 \Omega$
	$V = \frac{A}{A + R} E$ $V = \frac{1500}{1500 + 1612} (5)$
	$V = 2.41 \text{ V}$
(iii)	With internal resistance, the effective resistance of the circuit increases / terminal p.d. decreases.
	For the same change in temperature (same change in thermistor's resistance), there is a smaller change in the voltmeter reading.

Anglo-Chinese Junior College

H2 (9749) Physics

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